

AVC, parameters adaption analysis and comparisons

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1 What the measurement contains

Write the plant response, the disturbance and the control as complex numbers, $P = x_1 + ix_2$, $\Lambda = x_3 + ix_4$, $\vartheta = \theta_c + i\theta_s$. The algorithm's control law (its eq. 11) is then simply $\vartheta = -\hat{\Lambda}/\hat{P}$: the estimated disturbance divided by the estimated plant.

The algorithm models the measured residual as a sinusoid at the tracked frequency,

$$y(t) = \operatorname{Re}(Z) \cos \alpha(t) + \operatorname{Im}(Z) \sin \alpha(t), \quad Z = \vartheta \overline{P_\star} + \Lambda_\star = \Lambda_\star - \kappa \hat{\Lambda}, \quad \kappa = \overline{\left(\frac{P_\star}{\hat{P}}\right)}, \quad (1)$$

where $P_\star, \Lambda_\star$ are the true plant and disturbance.

Once the phase $\alpha(t)$ is known, the measurement therefore depends on the unknown parameters *only* through the complex number Z — two real numbers. This is not a sampling-rate statement: a 47 Hz tone sampled at 1 kHz gives ~ 21 points per period, but they are 21 noisy views of the same two numbers. Nor does it improve with time: since the regressor is $\mathbf{W} = \cos \alpha \mathbf{u}_1 + \sin \alpha \mathbf{u}_2$ with $\mathbf{u}_{1,2}$ phase-independent, $\sum_t \mathbf{W}\mathbf{W}^\top$ has rank two for any number of samples (verified over several hundred periods). More data buys precision on those two numbers, never additional rank. So the whole experiment yields two real numbers about the static parameters, and the question of which parameters to adapt is the question of which of them are determined by Z .

2 Three groups, three different answers

(a) The plant (x_1, x_2) : never observable

Treating (P, Λ) as jointly unknown, the map $(P, \Lambda) \mapsto Z$ is from $\mathbb{R}^4$ to $\mathbb{R}^2$ and has a two-dimensional kernel,

$$\delta Z = \vartheta \delta \overline{P} + \delta \Lambda = 0 \quad \Longleftrightarrow \quad \delta \Lambda = -\vartheta \delta \overline{P}. \quad (2)$$

Any change in the plant estimate can be exactly absorbed by a change in the disturbance estimate, leaving every measurement unchanged at every phase, for all time. The ambiguity is physical, not numerical: an experiment whose entire yield is one complex number cannot determine two complex unknowns, however long it is run. It cannot be removed by descending the cost more accurately, nor by enriching the disturbance; it would require an independent, known excitation of the command, which is precisely what an offline probe measurement does.

The plant is not merely hard to estimate—it is *not needed* as an estimate. By (1) only κ enters, and only through the sign of the feedback: the loop of Sec. (b) converges iff $\operatorname{Re} \kappa > 0$, i.e.

$$|\arg P_\star - \arg \hat{P}| < 90^\circ. \quad (3)$$

A magnitude error in $\hat{P}$ does not degrade the steady state at all: it only rescales the converged $\hat{\Lambda}$. What the algorithm needs from the plant is a phase, to within a very generous margin — a calibration constant, not a state.

(b) The disturbance quadratures (x_3, x_4) : observable once the plant is fixed

Hold $\hat{P}$ fixed. Then (1) is an affine bijection between $\hat{\Lambda}$ and Z (slope $-\kappa \neq 0$): two unknowns, two measurements, exactly determined. Restricting the adaptation to (x_3, x_4) is therefore not a workaround for the degeneracy of (2) — it is exactly the well-posed sub-problem that remains once the unobservable directions are excluded.

The update that the algorithm performs on these components is, after averaging over a cycle,

$$\dot{\hat{\Lambda}} \propto Z = \Lambda_\star - \kappa \hat{\Lambda}, \quad (4)$$

a first-order integral loop whose fixed point $\hat{\Lambda}_\infty = \Lambda_\star/\kappa$ gives $Z = 0$: *exact* cancellation of the tone, for any admissible $\hat{P}$. Note that $\hat{\Lambda}$ converges to whatever command cancels the disturbance, not to the disturbance itself; it is a controller state rather than an estimate. (Measured over three runs with different $\hat{P}$, the converged $|\hat{\Lambda}|$ tracks $|\hat{P}|$ to $\sim 1\%$, as $\hat{\Lambda}_\infty = \Lambda_\star/\kappa$ requires, while all three cancel equally well.)

(c) The frequency (ω, α) : observable, and necessary

A frequency error does not change the *value* of Z within a cycle; it makes Z *rotate* from cycle to cycle, at rate $\omega_\star - \hat{\omega}$. That rotation lies in the observable projection, not in the kernel (2), and it is what the phase detector $\langle y \sin \alpha \rangle$ of the PLL senses. Frequency is a property of the observed signal itself, recoverable from the output alone; the plant, by contrast, is a property of an input–output map that is never independently excited.

Adaptation of ω is also *necessary*. The pair (x_3, x_4) integrating in the frame rotating at α is a resonant (internal model) controller: its infinite loop gain sits exactly at $\hat{\omega}$ and nowhere else. If $\hat{\omega} \neq \omega_\star$, the integrand rotates and averages towards zero, and the correction collapses.

Summary

parameters	status	treatment
ω, α	observable (rotation of Z), and required	adapt (PLL)
x_3, x_4	observable once $\hat{P}$ is fixed; exactly determined	adapt (integral)
x_1, x_2	in the kernel; only $\arg \hat{P}$ matters, to 90°	calibrate, freeze

3 Experimental test

Closed-loop single-conjugate AO simulation (64px pupil, 8×8 Shack–Hartmann, 40 modes, 1 kHz, integrator gain 0.5 with 2-frame delay, 80 Hz actuator low-pass), with a 120 nm tip vibration at 47 Hz that the loop *amplifies* to a 890 nm residual. AVC acts on mode 0 in parallel with the integrator. Five 5 s runs share the atmospheric seed and vibration realisation and differ only in which parameters are adapted; the plant seed, where used, is the measured closed-loop value. Statistics over the second half of each run.

The table matches the analysis line by line.

- **Adapting x_1, x_2 destroys cancellation** (+0.0 dB), even though this run starts from the exactly measured plant. The unobservable directions (2) are free to drift, $\arg \hat{P}$ wanders, the margin (3) is violated and the integral loop of Sec. (b) reverses sign.
- **Not adapting ω destroys cancellation too** (+0.2 dB) when the frequency is held 2 Hz away from the truth — a large error for a resonant controller whose gain is concentrated at $\hat{\omega}$.

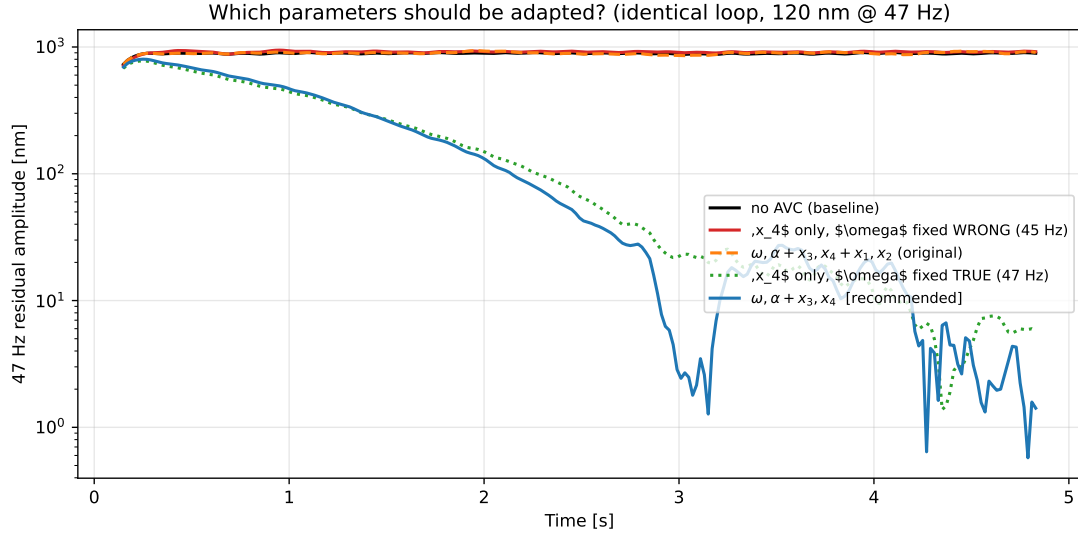


Figure 1: 47 Hz residual amplitude (300 ms lock-in). The two failing configurations sit on the baseline; the two working ones fall by more than a decade.

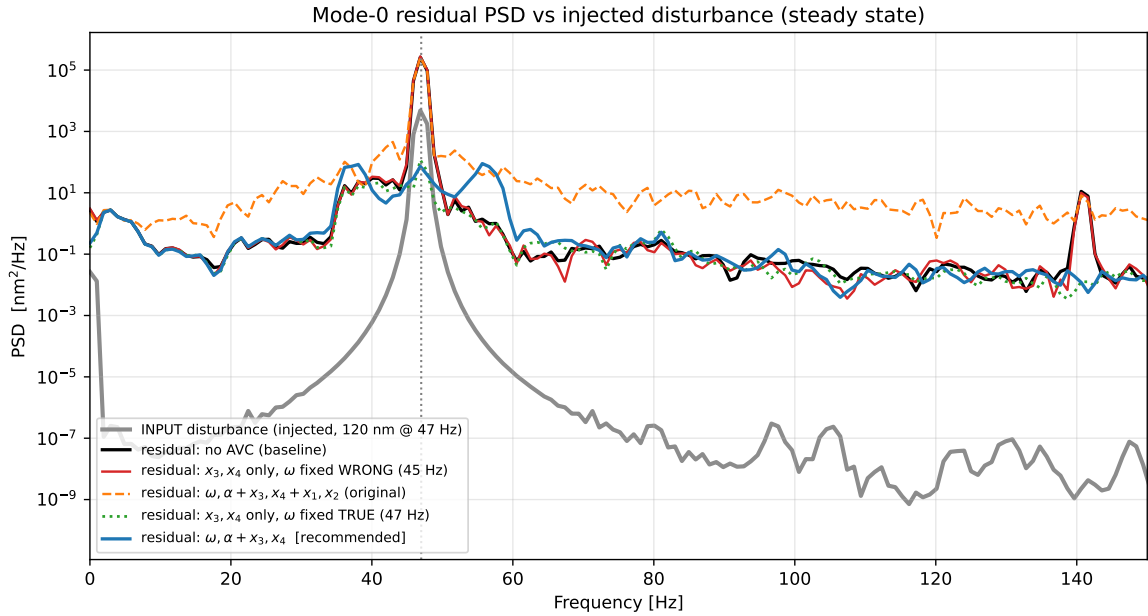


Figure 2: Steady-state mode-0 residual PSD for each configuration, together with the PSD of the injected disturbance (grey). The loop *amplifies* the tone: without AVC the residual peak stands +17.4 dB *above* the input. The two failing configurations track the baseline peak exactly; the two working ones remove it entirely, leaving the residual -18.2 dB *below* the input — a 35.6 dB swing at the tone. Note also the raised broadband floor of the original update (orange), ~ 1 –2 decades above every other configuration across the whole band: an unusable, wrongly-phased correction is being injected at all frequencies.

adapted parameters	mode-0 RMS	47 Hz amp.	vs. base	PSD vs. input	final $\hat{f}$
none (no AVC)	629.8 nm	890.3 nm	—	+17.4 dB	—
x_3, x_4 only, ω fixed <i>wrong</i> (45 Hz)	646.2 nm	913.6 nm	+0.2 dB	+17.6 dB	45.00
$\omega, \alpha + x_3, x_4 + x_1, x_2$ (original)	634.8 nm	892.6 nm	+0.0 dB	+17.4 dB	46.95
x_3, x_4 only, ω fixed <i>true</i> (47 Hz)	20.7 nm	16.7 nm	−34.5 dB	−16.1 dB	47.00
$\omega, \alpha + x_3, x_4$ [recommended]	26.1 nm	12.1 nm	−37.4 dB	−18.2 dB	47.00

Table 1: Ablation over the adapted parameter set. Both departures from the recommended set — removing frequency adaptation, or adding plant adaptation — destroy cancellation completely. The last-but-one column gives the residual PSD at 47 Hz relative to that of the injected disturbance (Fig. 2): the uncontrolled loop *amplifies* the tone by 17.4 dB, the recommended set puts it 18.2 dB below the input.

- **Holding ω at the true value works** (−34.5 dB), confirming that the PLL’s only job is to find ω ; and the PLL started 2 Hz away recovers that performance and slightly exceeds it (−37.4 dB), having converged to 47.00 Hz.
- **The spectra separate the two failure modes** (Fig. 2). Fixing ω at the wrong value leaves the residual spectrum indistinguishable from the baseline: the resonant controller is simply tuned elsewhere and does nothing. Adapting x_1, x_2 , by contrast, leaves the 47 Hz peak untouched *and* lifts the broadband floor by one to two decades across the whole band — the wandering plant estimate produces a large, wrongly-phased command that injects noise everywhere without cancelling anything. The second failure is therefore strictly worse than doing nothing.

4 Conclusion

The three parameter groups of the AVC algorithm are not interchangeable. The frequency is observable from the time evolution of the measured phasor and must be adapted, because the correction is a resonant controller tuned to it. The disturbance quadratures are exactly determined once the plant is held fixed, and their update is the integral action that performs the cancellation. The plant lies entirely in the unobservable kernel: it cannot be identified from a single tone, it is not needed beyond a phase accurate to 90° , and adapting it online is not merely futile but actively destabilising. Adapt (ω, α) and (x_3, x_4) ; calibrate the plant and leave it alone.

Reproducibility. Runs generated from `params_scao_avc_full_demo.yml` with the `gomega`, `freq` and `adapt_plant` keys of the `avc` block set as tabulated, against `params_scao_vibration_baseline.yml`. Companion reports: `avc_hybrid_note.pdf` (exact differentiation and why the cost is not a control objective) and `avc_exact_gradient.pdf` (full analysis).